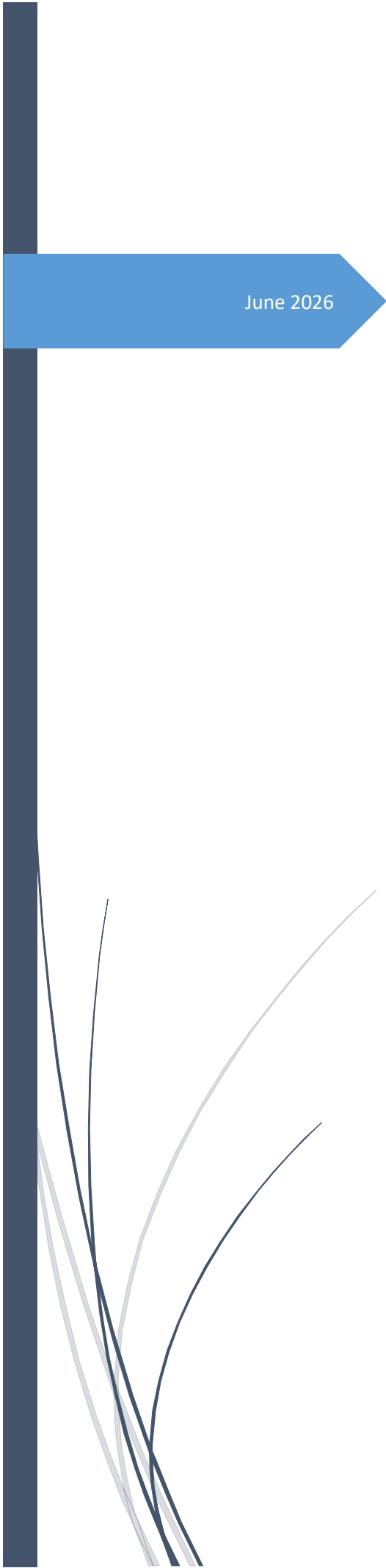


A dark blue vertical bar runs along the left edge of the page. A blue arrow-shaped graphic points to the right, containing the date. In the bottom left corner, several thin, curved lines in dark blue and light grey sweep upwards and to the right.

June 2026

PHISHING EMAIL DETECTION & AWARENESS REPORT

FUTURE INTERNS - CYBER SECURITY TASK 2

SEDO Kokou Henoc
FUTURE INTERNS

TABLE OF CONTENTS

- 1. **Executive Summary** Page 2
- 2. **Technical Tools & Environment** Page 2
- 3. **Practical Case Studies & Threat Analysis** Page 3
 - o 3.1 *Case Study 1: Spoofed Corporate Brand (High Risk)*
 - o 3.2 *Case Study 2: External Invoice Deception (Medium Risk)*
- 4. **Threat Classification Matrix** Page 4
- 5. **Security Awareness Guidelines for Employees** Page 5

1. Executive Summary

This report focuses on analyzing email security infrastructure to detect corporate phishing attempts. By analyzing raw email headers and identifying key social engineering indicators, this document serves as an official technical guide to training organization users against credential harvesting and malicious attachments.

2. Technical Tools & Environment

The following tools were utilized to perform threat analysis and simulation:

- **Google MessageHeader / MXToolbox Header Analyzer:** Used to extract and parse raw email metadata (hops, SPF, DKIM, DMARC signatures).
- **Browser Developer Tools:** Used to inspect embedded hyperlinks without executing malicious payloads.
- **Microsoft Word / PDF:** For client-ready technical documentation.

3. Practical Case Studies & Threat Analysis

3.1 Case Study 1: Spoofed Corporate Brand (High Risk)

- **Sender Display Name:** Microsoft Security Team
- **Visible From Address:** no-reply@microsoft-security-alert-hub.com
- **Target Link Destination:** [http://login.microsoft.security-theft.xyz/auth]
(http://login.microsoft.security-theft.xyz/auth)
- **Psychological Context:** High Urgency ("*Your account will be locked within 2 hours*").

Technical Header Analysis

Below is the extraction of the security headers parsed during the investigation:\

```
Delivered-To: employee@company.com

Received: from mail.fake-server.com

Authentication-Results: mx.google.com;

    spf=fail (google.com: domain of no-reply@microsoft-
security-alert-hub.com does not designate authorized IP
addresses)

    dkim=fail (signature verification failed)
```

Collez l'en-tête d'e-mail ou glissez-déposez le fichier texte ici

Veuillez envoyer un en-tête SMTP valide

[ANALYSER L'EN-TÊTE CI-DESSUS](#)

Aide

[Comment récupérer les en-têtes d'e-mails ?](#)
[Interpréter les en-têtes d'e-mails](#)

Que peut faire cet outil à partir des en-têtes

- Identifier les retards de distribution.
- Identifier la source approximative du
- Identifier le responsable potentiel.

Exemple de sortie

Subject:	Meetups this week with: Board gamers	Ce site utilise des cookies provenant de Google afin de fournir ses services et d'analyser le trafic.
SPF:	pass	
		En savoir plus. OK.

Key Findings:

1. **Authentication Failure:** The email header shows that SPF, DKIM, and DMARC all returned a **FAIL** status. This proves the email did not originate from official Microsoft servers.
2. **Domain Typosquatting:** The link uses the words "microsoft" and "security" but the actual root domain is `security-theft.xyz`, which is a malicious external server.

3.2 Case Study 2: External Invoice Deception (Medium Risk)

- **Sender:** `accounting@external-vendor-payments.net`
- **Attachment Name:** `Invoice_Pending_2026.pdf.exe`
- **Classification:** ● Suspicious

Key Findings:

1. **Double File Extension:** The attachment attempts to look like a PDF but contains a hidden `.exe` (executable) extension, indicating a malware deployment attempt.

4. Threat Classification Matrix

Assessment	Risk Level	Threat Indicators	Core Remediation
Verified Corporate Domains	Safe	Valid SPF/DKIM/DMARC tokens, expected business workflow.	No action required. Process email.
Lookalike External Domains	Suspicious	Tone changes, generic greetings, double-extension files.	Quarantine attachment. Contact IT Support.
Spoofed Identity & Fake Links	Phishing	Failed authentication headers, emergency tones, credentials harvesting URLs.	Block sender. Delete and flag to Security Operations.

5. Security Awareness Guidelines for Employees

1. **The Hover Check:** Users must hover their cursors over any link to inspect the actual destination before executing a click.
2. **Verify Out-Of-Band:** If an email from an executive or vendor requests urgent financial changes, verify via a known phone number or internal chat (Slack/Teams), never by replying to the e-mail.

